

Fish Species Prediction

Predicting fish species from physical measurements — benchmarking four classification algorithms with rigorous feature selection.

Logistic Regression

Decision Tree

Random Forest

XGBoost

Objective & Dataset

PRIMARY GOAL

Evaluate multiple classifiers, select a champion model, and identify key morphological predictors for species separation.

PIPELINE WORKFLOW

- 1 Preprocess — Standardize numeric features
- 2 Train & tune 4 candidate models via 5-fold CV
- 3 Evaluate on held-out test data partition
- 4 Extract feature importances and coefficients

DATASET PROFILE

159

Total Observations

7

Fish Species

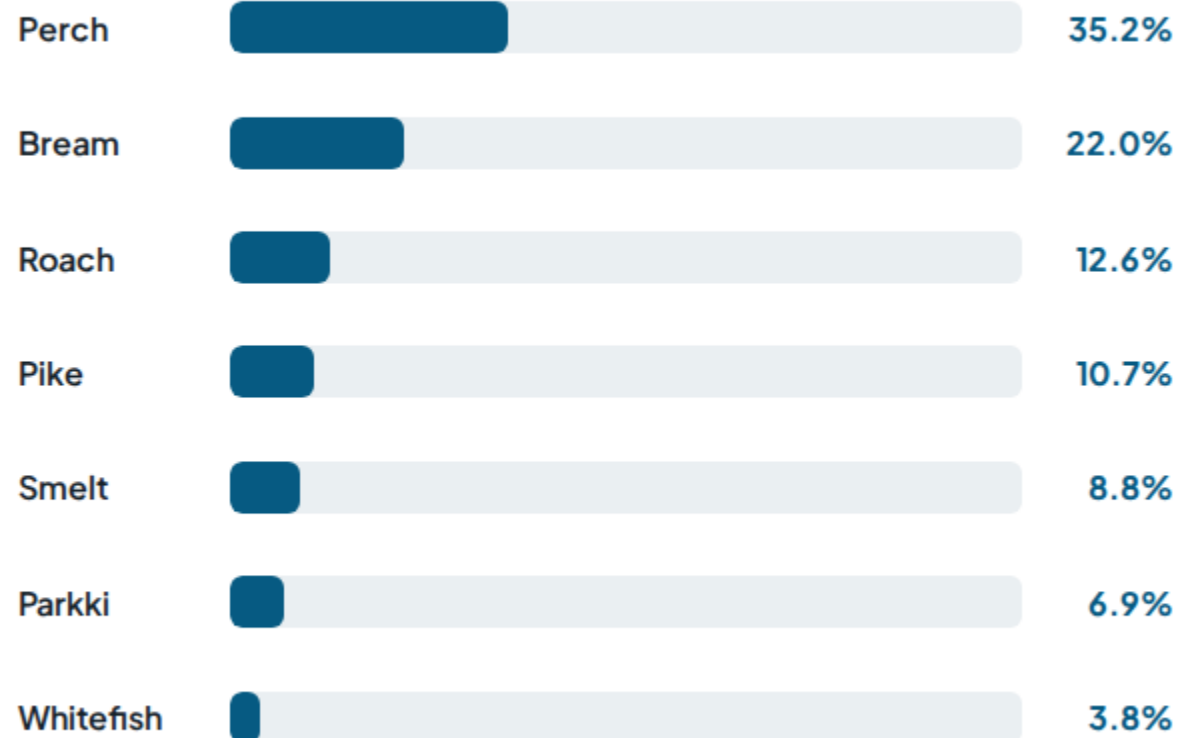
INPUT FEATURES

Weight, Length1, Length2, Length3, Height, Width

TARGET SPECIES

Bream, Parkki, Perch, Pike, Roach, Smelt, Whitefish

Species Distribution — Class Imbalance



KEY FINDING

Whitefish (~6 rows) and Parkki (~11 rows) are severely underrepresented.

This acute scarcity drives high metric volatility during validation, particularly in per-class precision and recall.

Fixing Multicollinearity

Length1, Length2, and Length3 were correlated at 0.99–1.00 — essentially duplicated signal.

Feature	VIF — Before	VIF — After
Weight	8.7	8.4
Length (1/2/3 → avg)	430 – 2085	7.7
Height	15.6	2.8
Width	12.4	7.0

** VIF > 10 signals problematic collinearity; VIF > 100 represents severe degradation.*

THE SOLUTION

Synthesized Length1, Length2, and Length3 into a single feature: length_avg.

All Variance Inflation Factors dropped strictly below 10. Coefficients stabilized completely, providing trustworthy interpretability for linear models.

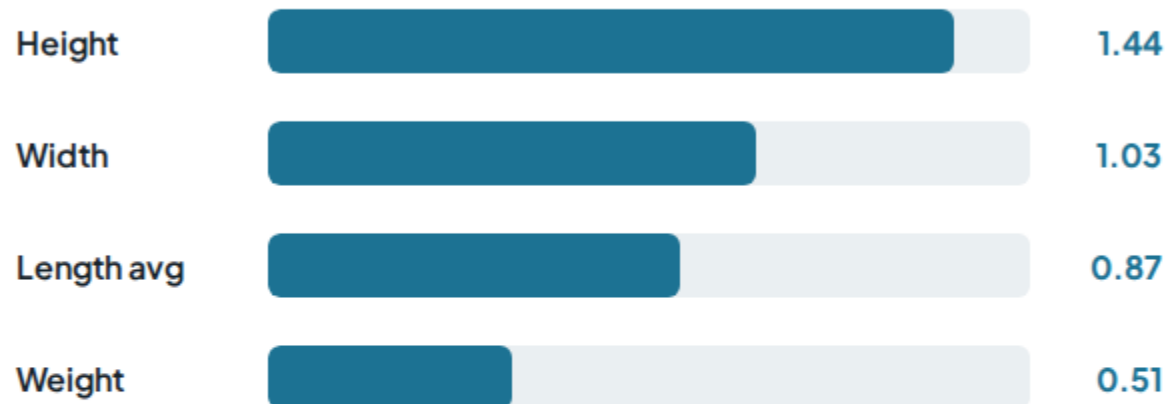
Model Performance Comparison

Model	Accuracy	Precision	Recall	F1 Score
Logistic Regression ★ Champion	0.813	0.708	0.813	0.752
Tuned Decision Tree	0.716	0.708	0.716	0.696
Tuned Random Forest	0.747	0.732	0.747	0.728
Tuned XGBoost	0.740	0.710	0.740	0.715

CHAMPION:

Logistic Regression outperformed tree-based models across accuracy (0.813), recall (0.813), and overall F1 (0.752) on the unseen holdout partition.

Feature Importance — Champion Model



Values represent mean absolute standardized coefficients ($|\beta|$) across all 7 species in Logistic Regression.

PRIMARY TAKEAWAY

Height is the dominant discriminator, closely followed by Width.

Geometric proportions and body ratios separate species far more effectively than overall mass (Weight) alone.

Limitations & Key Learnings

1 Cleanly Separable

Bream, Parkki, and Pike show near-perfect precision and recall consistently across all four candidate models.

2 Overlapping Classes

Perch, Roach, Smelt, and Whitefish overlap significantly in body dimensions, causing per-class metrics to fluctuate.

3 Macro vs. Weighted F1

Weighted F1 is heavily dominated by majority classes; macro-F1 uncovers hidden weaknesses in sparse minority classes.

4 Multicollinearity Fixed

Length feature VIF dropped from 2,085 to under 10, producing reliable, trustworthy model coefficients.

NEXT STEPS: Collect additional Roach & Whitefish samples, test SMOTE oversampling strategies, and optimize loss functions for macro-F1.